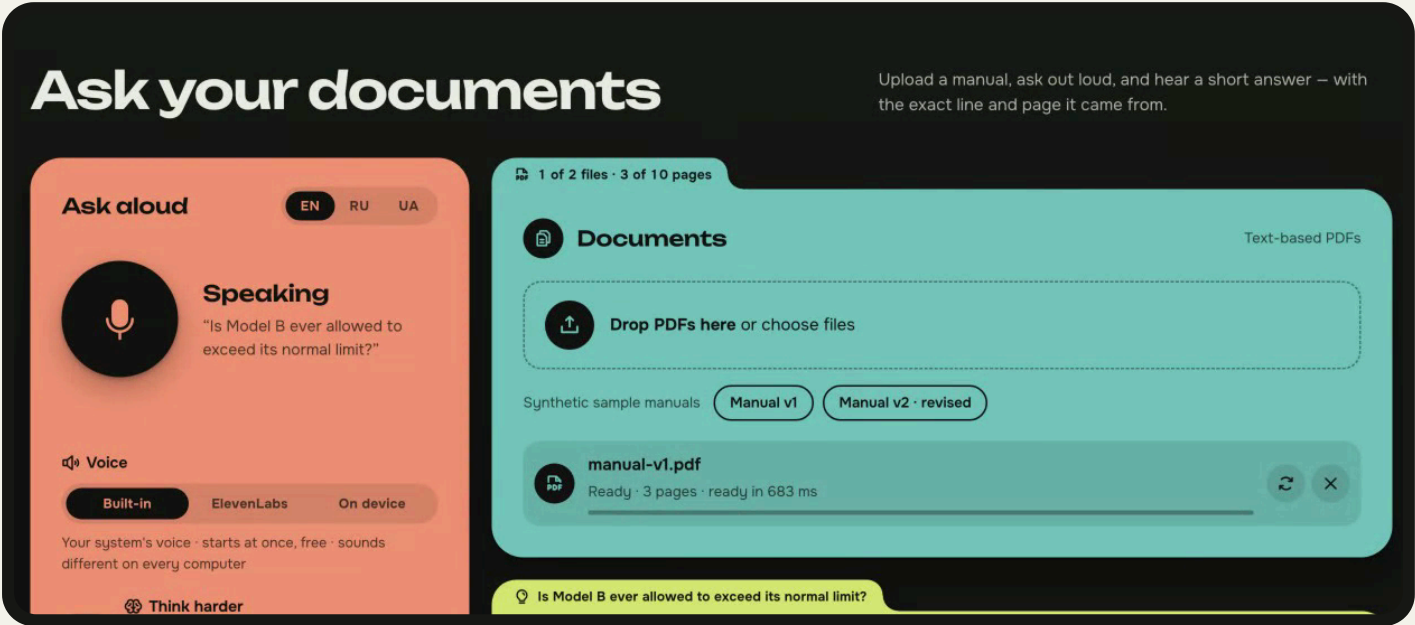
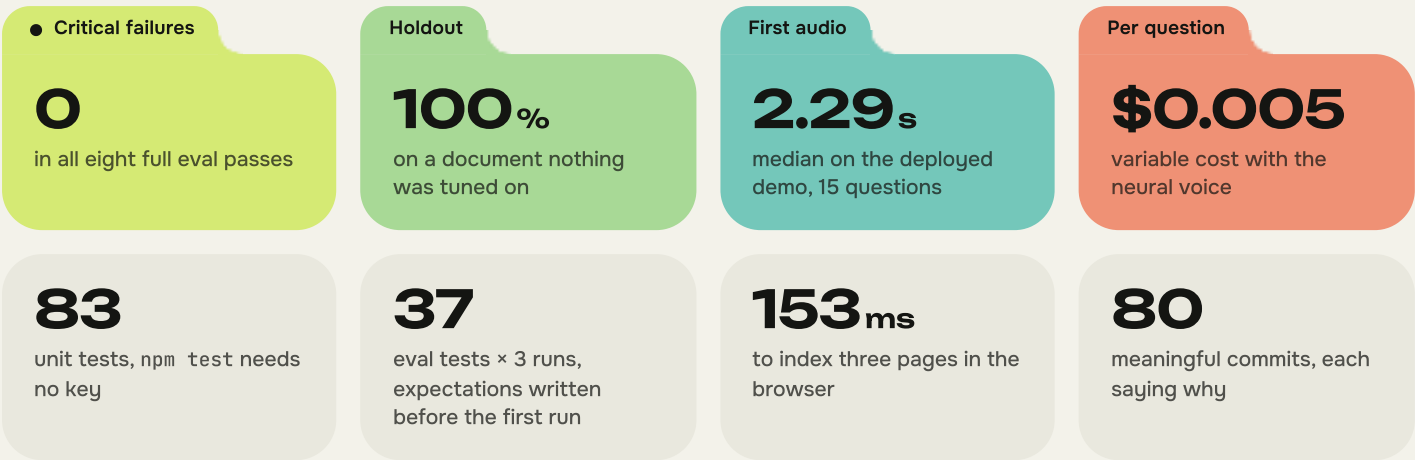


Ask your documents by voice

A full report on the prototype: what the brief asked for and how each point was met, how well, with which tools and why that way – plus everything built beyond the assignment, and the details you don't see at first glance.



The live app: the voice tile, documents, an answer "inferred from the document" with its WHY line, quote sheets and the measurements panel. A frame from a screen recording, not a mockup.

DEMO
askyourdocuments.vercel.app

REPOSITORY
github.com/AlekVoyan/test-CB

DATE
September 2026

What was asked — and what came out

The brief: a user uploads a manual and asks out loud; the service answers briefly by voice and shows the verbatim quote with its page. The answer comes only from the document, and replacing the document changes the answer. Up to two text PDFs, up to ten pages, one language, follow-up questions and an explicit "the document doesn't say".

● Verdict

The brief is met, parts of it with a wide margin

The core scenario — voice → answer → quote with page — works end to end, both locally and on the deployed demo. All six question types from the brief pass 3/3. Not a single plausible but unsupported answer reached the user in any run: that is the failure the brief names, and it is held off by code rather than by hoping the model behaves.

The strongest evidence is the **100% holdout**: a separate air-purifier manual the prompt was never tuned on. It answers the brief's point directly — "the app handles new input, not prepared answers".

Honestly

What isn't perfect

- **A6** — reasoning from the exception — is unstable: 1/3 and 3/3 in the last two passes with no code change. The decline is safe, but there is no answer.
- Every number was measured on NVIDIA's free endpoint, with a wide spread and 429 rate limits; the target, Claude Haiku 4.5, is waiting for a balance top-up.
- More than the suggested eight hours were spent — the breakdown is below.

Time

4.5_h

on the brief itself: extraction, retrieval, answer contract, validator, eval, voice flow, UI, measurements — within the eight hours

~16_h

beyond the brief, by my own choice: visual system, three voices, question understanding, languages, scans

20.8_h

in total by the log — 50 entries, each with its start and end

The brief suggests "up to eight focused hours". The core scenario was finished and verified within them; everything else is depth added on purpose, logged line by line in docs/TIME_LOG.md.

What the submission consists of

Working demo

Vercel Hobby: a static SPA and two serverless functions — /api/answer and /api/tts. Keys live only on the server.

Repository

README with a code map, .env.example, fixtures with their sources, eval with reports, delivery notes, time log, design system.

Video

A 2:55 script; nine scenes recorded with Playwright and cut in Remotion; two takes with a live microphone are shot by hand.

Every point of the brief — and what meets it

Requirement	Status	How it is met and what proves it
Ask by voice, hear a short answer	met	Web Speech API, the final transcript submits itself; a 1–2 sentence answer (≤ 45 words) is spoken in one of three voices. By voice in Chrome: end of speech $\rightarrow$ first audio, median 2.07 s.
Visible quote with its page	met	The model returns line IDs only; the quote text is filled in by code from the index — verbatim by construction. The page is on the sheet's tab, and "Open in page" opens the PDF itself with frames.
Answer only from the document	met	Validator: every number in the answer must be in the cited line or in the question; an unverified answer is never spoken. 0 critical failures in every pass.
Replacing the document changes the answer	met	M6: v1 $\rightarrow$ v2, 20 $\rightarrow$ 24 units, 3/3; the answer changed in 6/6 runs. The conversation resets when the documents change (A8).
Up to 2 PDFs, up to 10 pages, no OCR	met	Limits in config.ts; tests I1–I3 in npm test: a third file, 11 pages, a PDF with no text layer — a clear message, earlier documents stay.
Follow-up "and the other model?"	met	M3 and H3 3/3: the question is rewritten into a standalone one from the last four turns.
Say plainly when there is no answer	met	M5, A4, H5 3/3, zero citations. Beyond the brief — the decline suggests what the document does say nearby.
Fictional manual: 2 models, steps, limits, an exception	met	Kestrel Dosing System, Model A and B, the exception on p. 3, deliberate distractor numbers (12 V next to 12 units, 15 days next to 15 units).
Six questions + expectations and pages	exceeded	M1–M6 plus A1–A9, R1/R1b/R2, H1–H5, L1–L7, X1–X5: 37 tests. expected.json was committed before the first run — git is the proof.
Factual and citation scored separately; "plausible but unsupported" = failure	met	Two scores per run and a separate count of critical failures that never counts as success.
Ingestion separate from "question $\rightarrow$ first audio"	met	153 ms against 2.29 s in production; recognition / retrieval / model / validation / voice broken down right in the UI.
Cost: recognition, model, voice, retries, intermediaries; hosting separately	met	All five lines counted; free access priced at a paid provider's rate, as the brief asks; hosting \$0, listed separately.
Correction, ambiguity, refusing to infer	met	Correction (R2) — 3/3. Refusing to guess: M5, A4, X5 — 3/3, 0 invented answers. Ambiguity: code asks "which model?", and the spoken reply "Model B." is joined to the original question — R1 and R1b 3/3 (R1b used to range from 0/3 to 3/3).
AI tools and one example of checking their output	met	Claude Code (Opus 5), ChatGPT and Gemini for the spec draft; the check — the quote compared by hand with the rendered PDF page, and npm run verify-example.
Measurements over promises; what broke; time	met	All eight full passes reported, not the best one; anything unmeasured marked TBD; the time log line by line.
Working demo, repository, video ≤ 3 min	demo, code — yes	Demo and repository are ready; the video has a rough cut and is waiting for the live microphone takes.

37 tests, three runs each

Every session starts from an empty state and follows its steps: upload, ask, remove.
Scoring runs over the real PDFs through the same code as the app – expected answers are never fed into the pipeline.

● Reported pass · result per test

ID	Type	Question	Passed
M1	direct fact	What is the maximum for Model A?	3/3
M2	comparison	How do I set up Model A versus Model B?	3/3
M3	follow-up	And what about the other model?	3/3
M4	exception	Is Model B ever allowed to exceed its normal limit?	3/3
M5	absent fact	What is the battery life of Model A?	3/3
M6	after replacing v1 → v2	What is the maximum for Model A? → 24	3/3
R1 · R1b	ambiguity → spoken clarification	What is the limit? → Model B.	3/3 · 3/3
R2	correction	I meant Model B.	3/3
A1–A5	conflict, paraphrase, distractor numbers, unknown model, renamed file	anti-hardcode: v2 uploaded as manual-v1.pdf	15/15
A6–A9	reasoning from an exception, recognition error ("model bee"), history reset, document overview	Can Model B run at 15 units when it's 25°C?	12/12
H1–H5	holdout — an unseen document	room size, filters, follow-up, night mode, price	15/15
L1–L7	Russian and Ukrainian, switching language mid-session	А яке максимальне навантаження моделі В?	21/21
X1–X5	inference from a range, indirect fact, related line, slips	How often should I clean the nozzle?	12/15
I1–I3	input limits	3 files, 11 pages, a PDF with no text	pass

The spread — the honest headline

Six full passes in one day on the same code: P0 from **86.7%** to **96.7%** — temperature 0, identical prompts, only the sampling differs. The seventh and eighth, after the fixes in code: **100%**. **Every** pass is reported.

Holdout 100% in each. Critical failures 0 in each.

Pass	P0	P1	Holdout	Crit.
08:09	96.7%	96.3%	100%	0
08:20	90.0%	92.6%	100%	0
08:46	93.3%	100%	100%	0
11:17 · mishearing check	91.7%	96.3%	100%	0
13:55 · previously reported	86.7%	91.7%	100%	0
14:20 · same code*	90.6%	100%	100%	0
12a9764 · clarification in code	100%	92.6%	100%	0
8d4f0fd · reported	100%	100%	100%	0

* 41 of 111 questions came back 429 — an incomplete pass, not counted.

How one question travels

Documents, the index and the conversation live in the browser; the server is a stateless proxy that holds the key and builds the prompt itself. The same isomorphic code in `src/core` runs in the browser, on the server and in eval.

PDF → text
pdf.js, lines by coordinates, columns, hyphen joining

Index
line units with IDs like `d1:p2:s3` and frames on the page

Retrieval
BM25 with entity tokens like `model_a`; whole corpus or top-k

Model
structured output by JSON schema, zod, temperature 0

Validator
status and number rules, one retry with a hint

Quotes
text from the index by ID, checked again in the browser

Voice
Web Speech API recognition · spoken in one of three voices · first audio is measured at the audio output, not when the answer arrives

● Product stack

Interface	React 18 · Vite · TypeScript strict · Phosphor Icons
PDF	pdfjs-dist 6 · JBIG2 and JPEG 2000 decoders in WebAssembly
Contract	zod + JSON Schema for providers without zod
Model	Claude Haiku 4.5 (target) · NVIDIA Nemotron 3 Super (runtime)
Voice	Web Speech API · ElevenLabs Flash v2.5 · Supertonic 3 on ONNX Runtime Web + WebGPU
Hosting	Vercel Hobby, two functions
Tests	Vitest (83) · a custom eval harness

How I worked

Development	Claude Code on Claude Opus 5 – code, tests and docs under my direction
Spec draft	ChatGPT and Gemini, then rewritten with Claude Code
Design	the impeccable, emil-design-eng and animate skills; DESIGN.md as the contract
UI checks	Playwright: high-density frames, frame-by-frame animation review
Measurements	CDP, <code>performance.now()</code> , runs against production
Video	Playwright → ffmpeg → Remotion

Why two models

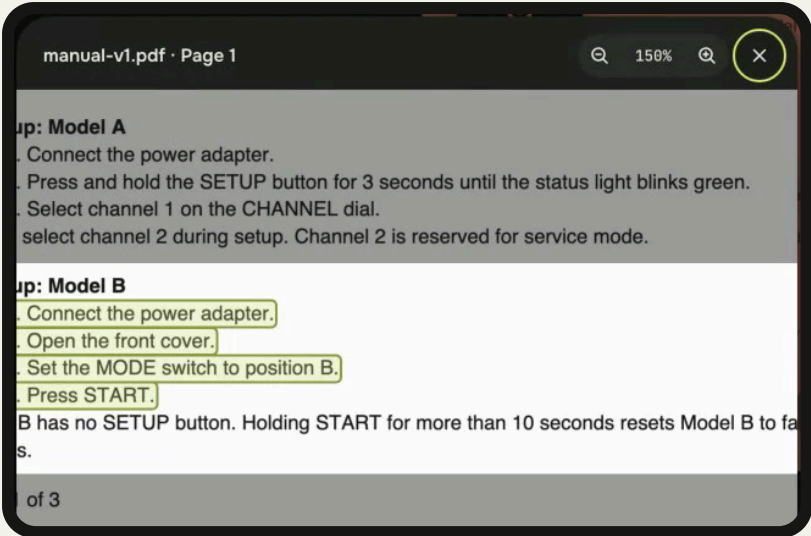
The Anthropic key turned out to have a zero balance within the first hour. The pipeline sat behind an `LLMClient` interface from the start, so a second adapter – NVIDIA Nemotron on the free endpoint – took one stage, not a rewrite. Its results are reported separately and priced at a paid provider's rate for the same model: "free credits are not zero cost", as the brief says. Switching back is one environment variable.

What was chosen — and why it beats the alternatives

Decision	Why	Why it beats the alternative
The model returns line IDs, code writes the quote	The quote is verbatim by construction: the model physically cannot paraphrase it.	Against "the model writes the quote": there verbatim is a promise, checked after the fact and sometimes broken.
Whole corpus in the prompt (full), top-k as fallback	10 pages is a few thousand tokens; no retrieval misses on paraphrases or on follow-ups with no keywords.	Against embeddings and a RAG framework: for 10 pages that is extra infrastructure and a new class of retrieval errors. Top-k switches on by itself when a document doesn't fit.
Validator + one retry, nothing unverified is spoken	The number rule catches a wrong quote for a right number; the retry gets a list of lines that contain the number.	Against "trust the model": on the free model the validator caught "12 units" cited to the wrong line — the user never saw it.
Documents in the browser, a stateless server	Serverless keeps no memory; demo viewers never see each other's documents; indexing costs \$0.	Against a server-side index: it would be lost between calls and would need storage and user isolation.
History resets when documents change	Otherwise "20 units" from v1 leaks into answers after switching to v2.	The cost — a follow-up doesn't survive a replacement, and that is tested (A8).
Analysis first, status second	With the status first, the model decided "not found" before reading and then justified it. A private analysis before the status fixed that.	Measured A/B, interleaved: ~34 extra tokens and no measurable delay.
The server builds the prompt	The client sends the question, history and lines — never prompt text.	The proxy can't be used as free model access; plus body-size limits and a rate limit.
Auto-submit the transcript	Faster; the transcript stays visible and editable, and a spoken correction "I meant Model B" works.	Against a confirm button: an extra click in a voice product. The cost — a misheard word goes straight through, hence the mishearing check.
No OCR, no embeddings	The brief excludes scans; 10 text pages need no vectors.	A scan without a text layer is rejected with a clear message; a scan with an OCR layer is accepted and answers.
Expectations committed before runs, holdout untouched	Git proves the correct answers were written before testing.	Against tuning to the test: twice a prompt fix made other tests worse — visible only because the test was written in advance.

A quote you can trust

A quote isn't a sentence next to the answer but a pointer into the source. You can open it on the PDF page itself and see where each line came from.



● Open in page

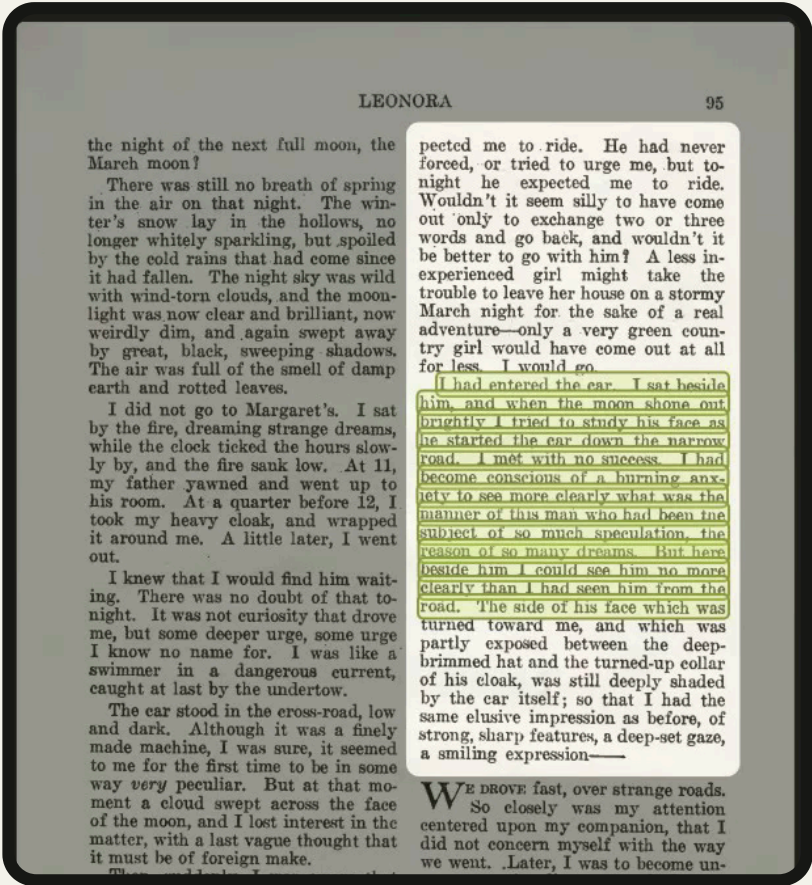
The panel renders **the PDF itself** with pdf.js, not our extracted text. The cited paragraph stays lit, the rest of the page sinks under a 42% veil, and every cited line is framed.

Zoom 100–300% with buttons, the keys + - 0, a pinch or Ctrl/⌘ with scroll – around the point under the pointer; the image scales at once and is redrawn sharp in an offscreen canvas when the gesture rests, so it never flashes.

Scans and two columns

A scanned 1927 magazine from the Internet Archive: each page is a JBIG2 image under an invisible OCR layer. Frames land inside the column the lines came from.

A column is found by an empty vertical band running the full height of a region; a label–value table stays whole, and a column may hold a nested pair of columns of its own.



Checked twice

The server checks the substring against the page, then the browser checks it again against its own local index. No match – the answer is withheld.

The number rule

A number in the answer must be in the cited line or in the question. Otherwise – a retry with the list of lines that contain it.

The paragraph sheet

Cited lines are in focus behind an ink rule, the lines around them fade under a mask: the sheet reads like a clipping from the page.

Three voices behind one switch

The product is a voice product, yet system voices sound robotic and differ on every computer: a reviewer on Windows may have no Russian or Ukrainian voice at all. So the voice is a choice, not an accident.

Built-in

System

The browser's speechSynthesis. Free, starts instantly — and stays the fallback for the other two: if the chosen voice can't speak, this one does, and the reason goes into the metrics.

● ElevenLabs

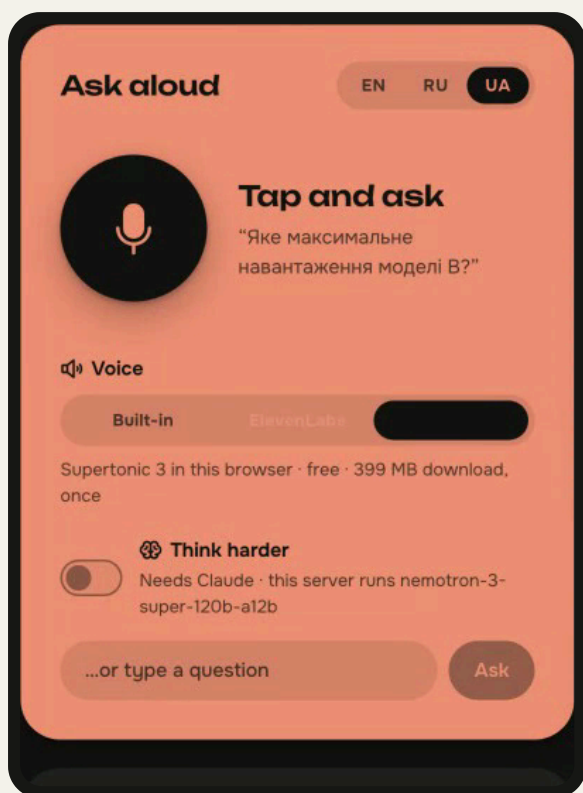
Neural

Flash v2.5, picked by listening against v3 and the macOS voices: first byte 208 ms against 749 ms, at half the price. PCM streamed into Web Audio, sound starts with the first chunk. ≈\$0.0046 per answer.

On device

Local and free

Supertonic 3 (99M) in a Web Worker on ONNX Runtime Web: WebGPU, with WebAssembly as the fallback. The model downloads once (399 MB) and lives in Cache Storage.



Protecting the paid voice

/api/tts speaks only text that /api/answer signed with an HMAC ticket, using a key derived from the API key. The public demo can't be used as a free TTS proxy.

The local voice in 0.75 s

The first version started speaking after 2.7–5.4 s: long sentences were synthesized whole, and every new input length made WebGPU build its kernels again. Now the first piece is short (up to 64 characters, cut at a sentence end, a comma or between words), each kind of piece has one fixed tensor shape warmed up at load, and a new answer cancels the old one between denoising steps. The result — **0.74–0.79 s** to first sound; WebGPU is ≈9× faster than WebAssembly.

Not obvious

- **Autoplay.** The browser allows sound only after a gesture, but the answer arrives seconds after the tap — so the AudioContext is primed right on the microphone press.
- **First audio is measured honestly:** the moment the sound reaches the output (including outputLatency), not when the answer arrived.
- **The voice language is pinned** (language_code) so Ukrainian text isn't read with a Russian accent.

When the question is loose and the answer isn't written down

Real manuals answer questions they don't contain word for word, and speech recognition hears the wrong words. Everything below works without a thinking mode, and nearly all of it without a single extra model call.

Q Is Model B ever allowed to exceed its normal limit?

Q Inferred from the document "Is Model B ever allowed to exceed its normal limit?"

Yes, Model B may exceed its normal limit of 12 units under specific conditions.

WHY The exception for Model B allows operation up to 15 units for no more than 5 minutes when ambient temperature is below 20°C.

III Speaking Using manual-v1.pdf · Model B

An inference from a rule, with its reason — shown and spoken.

Q What is the battery life of Model A?

Q Not in the document "What is the battery life of Model A?"

The uploaded documents do not specify the battery life of Model A.

Closest in this document

Setup: Model A Model A: the maximum load is 20 units. There is no exception for Model A

... Speaking Using manual-v1.pdf

A decline that suggests what the document says nearby.

● Inference with a reason

"Can Model B run at 40°C?" — not in the text, but the 5–35°C range is. The answer is marked *inferred* and carries a **WHY** line with the rule. A wrong inference counts as a **critical** failure — it is a new way to be wrong, so it is scored more strictly.

Mishearing check

A word that isn't in the document is folded phonetically (voiced/voiceless pairs, unstressed vowels, ph → f) and looked up among the document's words. One candidate — it is substituted and named in the answer. Two or more — the question comes back **without a model call**: \$0 and 0 ms.

The document's word

When there is no answer, the model may name the word the document uses for what was asked about. That word becomes the first suggestion — but only if it really appears in the document's lines; otherwise it is silently dropped.

Answer language

A Ukrainian question after a Russian one sometimes got a Russian answer. Now letters and frequent words are checked; a miss costs one retry — and if the retry misses too, the answer is kept: a correct answer in the wrong language beats "couldn't verify".

Questions from headings

Any uploaded document opens with three questions built from its own headings — instead of examples about someone else's manual.

A second retrieval pass

When a document doesn't fit the prompt, the best lines found feed their rare words back into the query — and the second search finds what the first missed.

Slips are named

If the model silently corrects "nozzel" to "nozzle", code notices and says so: "Assuming you meant "nozzle"".

The Proof Folder

A card file opened at night: each answer is a folder, and the line that proves it is filed behind it under a tab that carries the page. The tab's silhouette isn't decoration: a tab appears only when it carries data.

Answer
#d5ea74

Voice
#ef9175

Documents
#74c7ba

Proof
#a8d996

Filmic black
#161714

● The One Job Rule

Each pastel means one thing — in tiles, discs and latency chart segments alike: lime is the answer, coral the voice, teal the documents, sage the proof. Colour is never decoration. Black is lifted to a filmic #161714 with grain: there is no pure black anywhere.

Typography

Unbounded — a wide voice for names and headings only.
Onest — for the answer and everything that is read.
 JetBrains Mono — only where something was measured: timings, tokens, cost, line IDs. All three faces cover Cyrillic, including *і ї є ґ*.

Tab geometry

A tile is a rectangle with a 28 px radius. A folder's top-left corner is square, and above it sits the tab: a 14 px convex corner turns into a 14 px concave fillet. **The tab is exactly two radii tall** — so the edge reads as one continuous S-curve.

The tab has no height of its own: it spans from the row line to its folder's edge, and the corner and fillet are exactly half that distance. So the curve doesn't break on a shorter tab further back either. Depth and hover lift are two registered @property lengths: height, corners and edge are all computed from one interpolated value every frame — the gap between them is 0.01 px instead of 1.3 px.

p.2 · tab

28 = 2 × 14

convex corner → concave fillet
one continuous S-curve

Colour belongs to the plane

The front plane is full lime; each plane further back moves a quarter toward olive (0.086 in oklab lightness). Come forward — get lighter; step back — get darker.

Tabs don't jump

Every tab has its own place in the row. Bringing a folder forward changes only its depth, and the colour floods out from under its tab as a disc in 280 ms.

A shadow that follows the silhouette

The tab casts its shadow together with the tile, with no seam. Shadows are layered box-shadows, not a drop-shadow filter (more on that in the details).

Motion as meaning, not effect

Every movement answers "why": to file the answer into its folder, to show where a colour came from, to confirm a press. Anything repeated dozens of times is fast; what is seen once can afford some character.

Motion	Parameters	Why exactly this
Tiles enter	600 ms · 100 ms stagger · rise 200 px / 64 px on phones · quadratic ease-in-out	Once per load, so it is the one exception to the "under 300 ms" rule. Both halves are parabolas: an even speed-up and an even slow-down to a soft stop.
The answer files in	320 ms · rise 10 px · blur 3 px → 0	The blur glues the two states together, so the eye sees one transformation instead of a swap; quote sheets follow 50 ms apart.
A folder comes forward	disc from under the tab · 280 ms · cubic-bezier(0.23, 1, 0.32, 1)	Shows where the colour came from. The disc's scale is animated, not a clip-path – Chrome's compositor shifted the centre of a moving circle.
Hovering a folder	folder lifts · 160 ms	The whole folder rises – tab and edge together, so the silhouette never breaks. Fine pointers only: no false hovers on touch screens.
Press	scale 0.97 · 140 ms	Instant feedback: the interface "heard" the press.
Switches	indicator slides 220 ms · ease-out	Language, voice, "think harder": the selected option turns coral.
Microphone and speech	a ring every 1.6 s · three-bar equalizer	"Listening" and "speaking" read without words.
Page viewer	panel from the left on desktop · sheet from the bottom on phones	On desktop the answer and sheets stay visible on the right; Esc, the close button or the backdrop close it, and focus returns to the button.

● Reduced motion

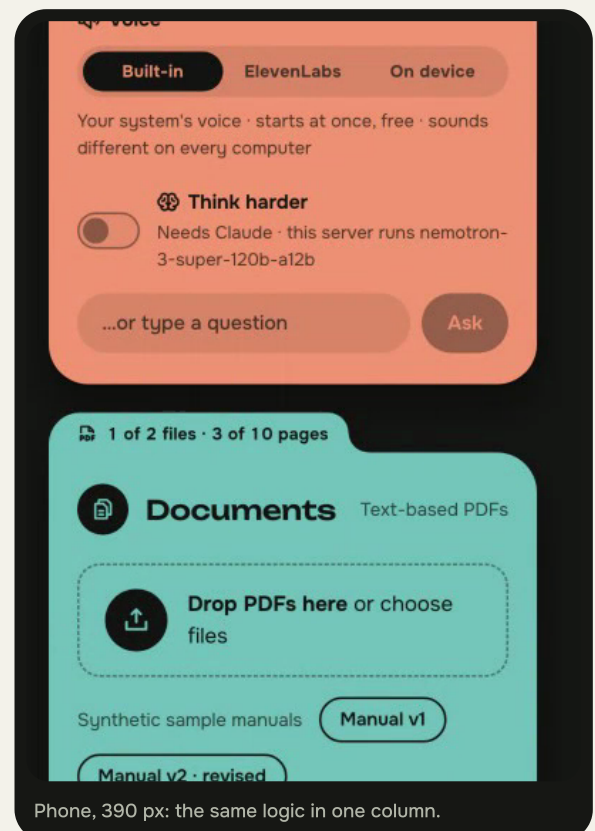
Every animation has a gentle variant: tiles only fade in, 40 ms apart, folders change depth in place, colour changes at once, the equalizer and the microphone ring stop. Not "zero motion", but "less and softer".

Grid

A 12-column bento up to 1280 px: the voice on the left across two rows, documents and the answer on the right, the answer as the largest tile. Below 1100 px – 6 columns; below 720 px – one column in reading order: voice → documents → answer → proof → measurements.

Browser surfaces

Text selection is lime with ink, the caret lime, scrollbars dark, focus always visible: 2 px with an offset, ink on pastels and lime on dark.



Decisions you don't see at first

01 • SHADOWS AND VOICE

Shadows stole the GPU from the voice

Tile shadows used a drop-shadow filter. A filter is recomputed every frame while anything inside or above it moves — the answer filing in, the equalizer playing. That took half the GPU from the local voice: synthesis took 1.5–2.1 s instead of 0.7 s. Found with a series of measured experiments; shadows became layered box-shadows.

02 • TEXT

A period in "1938r." split a fact

A line ending in an abbreviation was taken as a sentence end — the fact was torn from its date, and the validator rejected a correct answer because "the number isn't in the quote". An abbreviation no longer ends a line. After that, "What is a gas generator?" (asked in Russian) got its answer.

03 • IDENTIFIERS

Line IDs in any alphabet

Answering in Russian, the model sometimes wrote a line ID in Cyrillic — д2:p2:c16 instead of d2:p2:s16 — the line "wasn't found" and a correct answer was lost. Cyrillic look-alikes, ranges and a missing document prefix are now recognised. But the document is never guessed when two files could hold a line with that number.

04 • SCANS

Why the scan was blank

pdf.js decodes JBIG2 and JPEG 2000 with WebAssembly modules it fetches by URL rather than bundling. Without them the page image silently didn't draw, while the quotes were still exact — from the OCR layer. The decoders, CJK maps and fonts are now served under /pdfjs.

05 • MISHEARING

"Model C" will never become "Model B"

The mishearing check leaves single-letter identifiers and words after "model" alone: swapping in a neighbouring model is exactly a plausible but unsupported answer. Plus ≥6 letters, an everyday-word list and the same first sound: otherwise Russian "является" was substituted for "появляется".

06 • MEASUREMENTS

End of speech isn't always speechend

Chrome sometimes fires speechend after the final result, and recognition "took" 0 ms. End of speech is now speechend or the last interim result, whichever comes first.

07 • LANGUAGE

A retry doesn't throw away a good answer

The first language check spent the only retry on language; if the second answer broke another rule, the valid first one was lost too. Eval caught it. Now the answer set aside for its language is kept and returned if nothing better comes back.

08 • NUMBERS

A "clever" idea, rejected

I tried extending a citation automatically to the neighbouring line holding the number. Tests showed it could attach a line that doesn't support the claim. Reverted — the strict rule stayed, and the real fix was in text parsing (detail no. 2).

Twelve more details

09

The wheel only over tabs

The wheel scrolls the row of folders only over the tab strip and hands the page back at the ends – it never hijacks page scrolling.

10

Earlier answers are view-only

Clicking an old tab opens the old answer, but a follow-up always continues from the latest; "Back to latest" returns.

11

"Model bee"

Recognition spells the model's letter as a word. Code restores the letter after "model", and the question no longer turns into a clarification.

12

Suggestions without junk

A decline never suggests fragments from mid-sentence, figure captions or lists of model codes.

13

Related isn't proof

Lines related to a decline sit in dark folders with teal text, never in sage: sage is reserved for proof.

14

A reset you can see

When documents change, "Documents changed – conversation context reset" appears: the user knows why a follow-up didn't carry over.

15

Free ≠ zero

Nemotron on the free endpoint is priced at OpenRouter's rate for the same model: \$0.085 / \$0.40 per million tokens.

16

Cold start, honestly

The Vercel function was measured after 45 minutes idle: 1.12 s against ~0.4 s warm – the first question after a pause is 0.7 s slower.

17

Quotes by paragraph

A broad question about a one-page CV cited 26 lines in a single block. Now each paragraph is its own sheet, titled by its first line.

18

Columns in a CV

A label-value table isn't split into columns: there has to be running text on both sides of the cut.

19

Reports with reasons

The eval report keeps every rejected attempt and its reason – you see not only the result but why a retry was needed.

20

Heavy things on demand

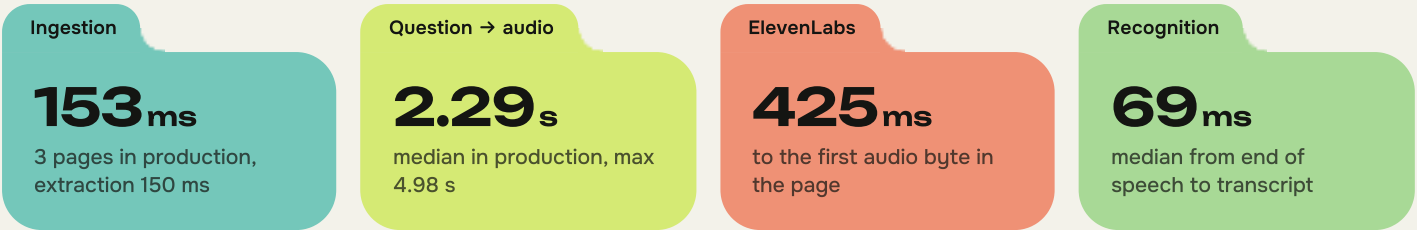
pdf.js decoders (3.9 MB) load only when a document needs them, and ONNX Runtime (25.7 MB) only if the local voice is chosen.

● The principle behind the details

Almost every detail here came about the same way: a real document or a real measurement showed a problem → the cause was found, not guessed → the fix was checked by a test or a run → if the fix broke something, it was reverted and written down as rejected. That is why the delivery notes have a "what didn't work" section – and it is more useful than the "what worked" one.

Measured, not promised

Every number comes from the real app. Nearly all of a question's time is the model call on the free endpoint; retrieval and validation take about a millisecond.



Measurement	Where	Result
Question → first audio, M1 / M4 / M5 × 5	Vercel + NVIDIA + ElevenLabs	2.42 / 2.65 / 2.11 s
End of speech → first audio, 8 spoken questions	Chrome 152	median 2.07 s
Local voice: answer → first piece synthesized	WebGPU on Metal	0.74–0.79 s
Local model: first download / from cache	headless Chromium	≈35 s / 4–6 s
Question → first audio, a not-found answer, one call	embedded Chromium	1.23 s
Function cold start after 45 min idle	Vercel	1.12 s

● Cost of one question

Ingestion	\$0
Model, Nemotron 3 Super (at a paid rate)	\$0.00024
Model, Claude Haiku 4.5 (estimate)	≈\$0.0029
ElevenLabs Flash v2.5 voice, ~92 characters	≈\$0.0046
On-device / built-in voice	\$0
Retries (1.8% of questions) – already included	included
Intermediaries	\$0
Total with the neural voice	≈\$0.0048

Hosting — separately

Vercel Hobby, \$0 a month within the plan's limits.

Voice is the main expense

The neural voice costs 18× the reasoning on Nemotron and 1.5× on Haiku. Hence the free local option.

Price sources

Every price in docs/pricing.md comes with a URL and the date it was checked.

What isn't solved — and what I'd do next

Known limitations

- **A6 is unstable.** On "Can Model B run at 15 units at 25°C?" the model sometimes reasons correctly but files it as "not found": 1/3 and 3/3 in the last two passes with no code change. The decline is safe.
- **Endpoint spread.** One pass on the free model is a weak measurement; by the sixth pass of the day the endpoint was returning 429s.
- **Haiku is unmeasured** — the Anthropic balance was empty; the adapter is written and covered by tests.
- Code-side "which model?" only recognises models written "Model X": for "Nimbus Pro" the model still decides.
- Suggestions under a decline aren't spoken: the paid voice says only server-signed text.
- On scanned magazines, "headings" can be title-page boilerplate.

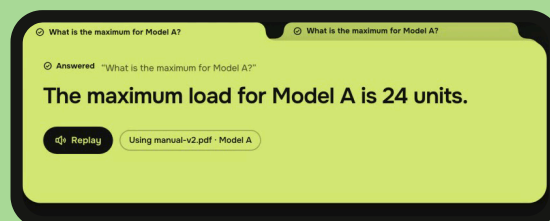
Next

- **A pass on Claude Haiku 4.5** and re-measuring A6 and X3 — first of all.
- **Streaming the answer:** speak the first sentence as soon as it passes validation.
- **Recognizer alternatives:** pick from Chrome's several hypotheses the one whose words are in the document.
- **Server-side recognition** for every browser, with temporary keys.
- **A rule-evaluation step** for "is it allowed" and range questions instead of more prompt text.
- **Documents across reloads** (IndexedDB) and a real rate limit on KV.

How AI output was checked

Automatically, on every answer: non-existent IDs, broken status rules, a number that isn't in the cited line — and the substring checked again in the browser.

By hand, one example end to end: the Model B exception answer opened with "Open in page"; the framed line on the rendered page compared with the quote word for word and number for number; the same question paraphrased — same line, same page; the distractors "15 days" and "no exception for Model A" stayed out of the answer. Repeatable with `npm run verify-example`.



What I wanted this prototype to show isn't "the model is smart" but **a system that doesn't let it lie** — and an honest account of where it still gets things wrong.